

```

1 "/Users/son/Library/Mobile Documents/com~apple~
  CloudDocs/University/Laboratory/Year 3 Lab/
  pycharm_venv/bin/python3" /Users/son/Library/Mobile
  Documents/com~apple~CloudDocs/University/
  Laboratory/Year 3 Lab/plotting.py
2 /Users/son/Library/Mobile Documents/com~apple~
  CloudDocs/University/Laboratory/Year 3 Lab/plotting
  .py:20: RuntimeWarning: invalid value encountered
  in divide
3   Q_errs = heats * np.sqrt((err_P/power)**2 + (
  err_times/times)**2) #Error in Heat
4 Plot saved at /Users/son/Library/Mobile Documents/
  com~apple~CloudDocs/University/Laboratory/Year 3
  Lab/Results/Total Intensity within the Region of
  Interest over Q.png
5 2025-01-21 19:03:09.039 python3[29732:54770447] +[
  IMKClient subclass]: chose IMKClient_Modern
6 2025-01-21 19:03:09.039 python3[29732:54770447] +[
  IMKInputSession subclass]: chose
  IMKInputSession_Modern
7 Plot saved at /Users/son/Library/Mobile Documents/
  com~apple~CloudDocs/University/Laboratory/Year 3
  Lab/Results/Signal Improvement [%] in Signal over Q
  .png
8 Plot saved at /Users/son/Library/Mobile Documents/
  com~apple~CloudDocs/University/Laboratory/Year 3
  Lab/Results/Signal-to-Noise Ratio within the Region
  of Interest over Q.png
9 Plot saved at /Users/son/Library/Mobile Documents/
  com~apple~CloudDocs/University/Laboratory/Year 3
  Lab/Results/SNR Improvement [%] in SNR over Q.png
10 Points where SNR goes from positive to negative: ['
  87.24687634640243 +/- 7.500136887886854 J', '108.
  186126669539 +/- 9.513397824103533 J', '303.
  6191296854804 +/- 36.61254101165438 J', '450.
  19388194743647 +/- 69.59325012768905 J']
11 Points where SNR goes from negative to positive: ['
  73.28737613097803 +/- 6.220642393123861 J', '101.
  20637656182681 +/- 8.828707478531218 J', '136.
  1051271003878 +/- 12.405432607757142 J', '310.
  5988797931926 +/- 37.9187844554916 J']

```

12

13 Process finished with exit code 0

14